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31 MARCH 1960
SUPERSEDING
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MILITARY SPECIFICATION

INSTALLATION KIT, GAS-PARTICULATE FILTER UNIT, COMBAT VEHICLE, M20

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of filter unit installation kit.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.** The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-P-268 — Paper, Kraft Wrapping.

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

PPP-C-843 — Cushioning Material, Cellulosic.

PPP-P-291 — Paperboard, Wrapping, Cushioning.

PPP-T-60 — Tape; Pressure-Sensitive Adhesive, Waterproof — For Packaging and Sealing.

PPP-T-76 — Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-B-117 — Bags, Interior, Packaging.

MIL-B-121 — Barrier-Material, Greaseproofed, Flexible (Waterproofed).

MIL-B-130 — Barrier-Material, Paper Noncorrosive.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

FSC 4240

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MIL–STD–129 — Marking for Shipment and Storage.

DRAWINGS

CHEMICAL CORPS

D5–19–1055 — Installation Kit, Gas Particulate Filter Unit, Armored Vehicle, M20 — Bill of Material.

B136–52–60 — Test Lamp.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33d Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components used in the manufacture and assembly of the installation kit shall conform to the specifications and drawings forming a part of this specification.

3.2 Manufacture. The installation kit shall be manufactured in accordance with the drawings and specifications listed on Drawing D5–19–1055.

3.3 Workmanship. Workmanship shall be of such nature that when proper procedure is followed the kit is free from damage, defects, or contamination which detrimentally affect the suitability of the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Objective evidence. The supplier shall present objective evidence as required by the Government representative that all materials and components are in accordance with 3.1 and section 5 (see 6.4).

4.3 Alternate inspection (including testing) procedures. The supplier may utilize any alternate inspection procedure which will assure equal or better quality by submitting a written proposal, with justification, and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of this specification will govern.

4.4 Inspection provisions.

4.4.1 *Regular production lot.* A regular production lot shall consist of the installation kits produced by one manufacturer, from the same materials, and under essentially the same manufacturing conditions.

4.4.2 *Sampling.* Sampling shall be conducted in accordance with Standard MIL–STD–105.

4.4.3 *Examination.* Sample installation kits shall be examined in accordance with the classification of defects and with Standard MIL–STD–105.

4.4.4 *Classification of defects.*

4.4.4.1 *Frame—assembly.* (Dwg. E5–19–1054 which is listed on Dwg. D5–19–1055.)

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Categories and defects

Major *AQL 1.0 percent defective*

1. Components missing, incorrect or incorrectly located or assembled.
2. Hardware components inoperative.
3. Welding incorrect or missing.
4. Alignment of mounting holes in frame with holes in shock mount assembly incorrect.
5. Damage (see 3.3).

Minor *AQL 2.5 percent defective*

51. Protective finish missing, inadequate or incorrect.

Incidental *AQL 10.0 percent defective*

101. Contamination.
102. Burrs or sharp edges.

4.4.4.2 *Cable and Harness Assemblies.* (Dwgs. B5–19–1057, C5–19–1064, C5–19–1065, B5–19–1066, B5–19–1067, B5–19–1068, and B5–1069, which are listed on Dwg. B5–19–1055.)

Categories and defects	*Acceptance standards*
Major A *AQL 0.65 percent defective*	
1a Electrical continuity	4.4.5
Major *AQL 1.0 percent defective*	
1. Component missing, incorrect or improperly assembled.	
2. Damage.	

4.4.4.3 *Lamp Switch Assembly.* (Dwg. C5–19–1059 which is listed on Dwg. D5–19–1055.)

Categories and defects	*Acceptance standards*
Major A *AQL 0.65 percent defective*	
1a Switch and lamp, nonfunctioning	4.4.5
Major *AQL 1.0 percent defective*	
1. Component missing, incorrect or improperly assembled.	
2. Damage.	

4.4.4.4 *Preparation for delivery.* (Section 5.)

Categories and defects

Major *AQL 2.5 percent defective*

1. Preservation on unpainted surface missing.
2. Greaseproof wrap on preserved surfaces missing.
3. Quantity per container incorrect.
4. Containers incorrect.
5. Unit packages and exterior container closure incorrect.
6. Marking, incorrect, illegible or missing.
7. Container damaged (broken, crushed, torn, punctured or wracked).
8. Strapping loose or missing.

Minor *AQL 4.0 percent defective*

51. Greaseproof envelope not as specified.
52. Insufficient cushioning or blocking to insure a tight package.
53. Metallic and non-metallic parts mixed.

Note. Inspect prior to packaging where applicable.

4.4.5 *Cable Continuity and Lamp Switch Functioning.* Determine cable continuity and lamp switch functioning using Chemical Corps Test Lamp, Dwg. B136–52–60.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and drying. Prior to preservative application as specified in 5.2.1 unpainted bare metal surfaces shall be cleaned and dried in conformance with Specification MIL–P–116, using method C–1, and any of the drying methods listed therein.

5.2 Preservation and packaging.

5.21 *Level A.* Levels of preservation and packaging shall be as specified (see 6.2).

5.2.1.1 Unit packaging.

5.2.1.1.1 *Frame assembly.* Spring catch of frame assembly shall be oiled with Type P–7

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preservative oil conforming to Specification MIL–P–116, actuating catch during application to assure penetration of the preservative oil, wiping off excess.

5.2.1.1.2 *Shock mount assembly.* The unpainted surfaces of the shock mount assembly shall be coated with Type P–1 preservative compound, conforming to Specification MIL–P–116, and overwrapped with type III noncorrosive barrier material conforming to Specification MIL–B–130.

5.2.1.1.3 *Bracket and strip assembly.* The threaded surfaces of these assemblies shall be coated with Type P–1 preservative compound, conforming to Specification MIL–P–116, and then cushion wrapped together with type III noncorrosive barrier material, conforming to Specification MIL–B–130, and placed together in one style 2 or 4 paperboard folding box conforming to Specification PPP–B–566. The box shall be closed and secured with minimum 1-inch wide pressure-sensitive tape, conforming to Specification PPP–T–76.

5.2.1.1.4 *Metal parts.* The metal parts listed below shall be coated with Type P–7 preservative oil, conforming to Specification MIL–P–116, and placed in non-heat sealed envelopes fabricated from type III, grade A, class 2 barrier-material, conforming to Specification MIL–B–121: the aluminum tube in one envelope; the bronze terminal in one envelope; the steel pads, bosses and nuts in one envelope, and the steel clamps, screws and washers in one envelope. Envelopes containing preserved metal parts shall then be placed in a snug-fitting style 4 paperboard folding box, conforming to Specification PPP–B–566, with carton closed and secured with minimum 1-inch wide pressure-sensitive tape, conforming to Specification PPP–T–76. The carton shall then be inserted in a type II, class b waterproof interior packaging bag, conforming to Specification MIL–B–117, with the bag heat-sealed.

	Quantity
(a) Tube, aluminum alloy, seamless 1¾" by 7.8" O.D.	1 ea.
(b) Terminal, crimped, female, bronze	1 ea.
(c) Pad, steel, ¼" by ½" dia.	24 ea.
(d) Boss, steel, ⅜" by ⅝" dia.	4 ea.
(e) Nut, steel	4 ea.
(f) Clamps, steel	2 ea.
(g) Screws, steel	12 ea.
(h) Washers, steel	8 ea.

5.2.1.1.5 *Non-metallic parts.* Non-metallic parts listed below shall be packaged together in accordance with method III of Specification MIL–P–116:

	Quantity
(a) Sleeve, plastic	1 ea.
(b) Shell, male, rubber	1 ea.
(c) Grommet	1 ea.
(d) Connector	1 ea.

5.2.1.1.6 *Lamp, switch, assembly.* No preservative shall be applied. Assembly shall be packaged in accordance with method IA–8 of Specification MIL–P–116.

5.2.1.1.7 *Electrical cable and electrical harness cable assemblies.* No preservative shall be applied. The exposed ends of all cable assemblies shall be covered with type I, class 1 pressure-sensitive tape, conforming to Specification PPP–T–60. Cable assemblies shall then be individually wrapped in grade B kraft wrapping paper, conforming to Specification UU–P–268. Prior to wrapping, all assemblies exceeding 1 foot in length shall be coiled.

5.2.1.1.8 *Drawings.* The drawings shall be packaged in accordance with method IC–3 of Specification MIL–P–116.

5.2.1.2 *Intermediate packaging.* Parts assemblies and drawings preserved and unit packaged as specified in 5.2.1.1 shall be intermediate packaged in a snug-fitting style

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RSC, W5c fiber box, conforming to Specification PPP-B-636, having approximate inside dimensions of 13½ inches by 7 inches by 9 inches. Internal cushioning or blocking materials, or both, conforming to Specifications PPP-C-843, PPP-B-636, or PPP-P-291 as applicable shall be used as required to make a tight pack. The fiber box shall then be closed by taping in accordance with the appendix to Specification PPP-B-636 for interior containers.

5.2.2 *Level C.* Preservation, as applicable, and packaging of assemblies and parts comprising the installation kit, including drawings, shall be in accordance with the manufacturer's commercial practice.

5.3 **Packing.** Levels of packing shall be as specified (see 6.2).

5.3.1 *Level A.* Installation kits, packaged as specified in 5.2, shall be packed not to exceed 200 pounds gross weight in a class 2, style 2 or 2½ nailed wood box, conforming to Specification PPP-B-621. The boxes shall be closed as specified in the applicable requirements of Specification PPP-B-621, with strapping to be in accordance with the appendix thereto, using galvanized strapping.

5.3.2 *Level B.* Installation kits, packaged as specified in 5.2, shall be packed as specified in 5.3.1, except that class 1, style 2 or 2½ nailed wood boxes and coated-finish strapping shall be used.

5.3.3 *Level C.* Installation kits, packaged as specified in 5.2, shall be packed in a manner to insure carrier acceptance and safe delivery at first destination for immediate use. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation. Containers used shall be of uniform size, shape and material.

5.4 **Marking.** In addition to any special markings required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 **Intended use.** This specification covers one type of installation kit designed to facilitate installation of M8A2 filter unit in various combat vehicles and is intended to be used with detailed Chemical Corps drawings for each vehicle, as listed below:

Vehicle nomenclature	*CmlC dwg. No.*
(a) Tank, 76-mm gun, M41 (T41E1)	D5-19-1272
(b) Tank, combat, full-tracked: 90-mm gun, M46	D5-19-1274
(c) Tank, combat, full-tracked: 90-mm gun, M47	D5-19-1268
(d) Tank, combat, full-tracked: 90-mm gun, M48	D5-19-1273
(e) Tank, recovery vehicle, medium: M74	D5-19-1267
(f) Vehicle, heavy recovery: T51	D5-19-1271
(g) Carrier, personnel, full-tracked: armored, M75	D5-19-1270
(h) Carrier, personnel, full-tracked: armored, M59	D5-19-1269

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of preservation, packaging, and packing.

6.3 **Stock number.** The Federal stock number for this item is 4240-606-7575.

6.4 **Objective evidence.** Provisions for objective evidence and inspection records, and maintenance of inspection records will be specified by the procuring activity.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in

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